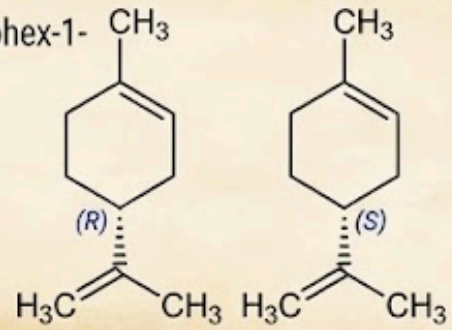
LIMONENE

IUPAC Name:
(R)-4-isopropenyl-1-methylcyclohex-1-

Molecular Formula:
$C_{10}H_{16}$

Molar Mass:
136.23 g/mol



Main component of **CITRUS ESSENTIAL OILS!** Responsible for the key aroma.



1. Structure and Identity

Common name: Limonene
Family: Monoterpene
Appearance: Clear, colorless to pale yellow liquid
Natural sources: Peels of oranges, lemons

2. ESSENTIAL PROPERTIES

- Strong citrus aroma
- Water-insoluble
- Volatile (easily vaporized)
- Exists as two optical isomers (chiral)

3. What Makes it Amazing?

Powerful Solvent

Potent and pleasant fragrance

Important in flavor chemistry
Natural insecticide properties



4. Creative and Industrial Uses

- Flavoring Agents
- Fragrance Ingredients
- Natural, bio-solvent
- Component in paints and coatings

5. Aroma and Benefits

- Aroma therapy (stress relief)
- Potential anti-inflammatory effects
- Study-supportive for cognitive function

6. Industrial & Green Chemistry

- Essential for the essential oil industry
- Used as a renewable, plant-derived solvent
- Research for biofuel precursors

APA Citations

- National Center for Biotechnology Information. (2024). PubChem compound summary for Limonene.
- National Institute of Standards and Technology. (n.d.). Limonene. NIST Chemistry WebBook, SRD 69.